Geology Rocks!: Practice Test 1

Time: 50 minutes. Partners rotate through 14 stations, about 3 and a half minutes each, and may not go back or skip ahead. You may use your one 1-inch team binder, pencils, and up to 2 hand lenses; the tools you need are at each station. Write the specimen name first, then answer the rest, and leave every specimen at its station.

Partner names: _____ and _____ Score: _____

Station 1: Rock or Mineral?

1. Which specimen is a rock: A, B, or C? How do you know?

2. Name specimens B and C.

3. B and C are both found inside A. Which is harder, B or C? Give both Mohs numbers.

4. Specimen A is the North Carolina state rock. Name one way people use it.

Station 2: Soft Minerals

5. Name specimens A, B, and C.

6. Which specimen is number 1 on the Mohs scale?

7. Match each specimen to a use: baby powder, drywall, electrical insulation.

8. Look at C (if it is a real piece, gently lift an edge with your fingernail). What property lets it split into thin sheets?

Station 3: Cube Minerals

9. Name specimens A, B, and C.

10. A and B both break into cube shapes. Without any tools, how can you tell B from A? (Use the clues on card B.)

11. Try to scratch C with the penny (about 3.5), then with the steel nail (about 6.5). Which one scratches it? About how hard is C?

12. Which specimen is the main ore of lead? Which one is spread on icy roads?

Station 4: Streak Test

13. Rub A on the streak plate. Write its streak color. Then read card B: what streak color does B leave?

14. Name specimens A and B.

15. C is a photo card. Gold is soft, very heavy, and bends instead of breaking. Name C and tell what color its streak would be.

16. Which specimen here is the main ore of iron, used to make steel?

Station 5: Metal and Pencil

17. Name specimens A and B.

18. Rub B gently on the paper. What happens? What school item uses B?

19. Why is A used to make electric wires?

20. Describe the luster (shine) of A.

Station 6: Hard Minerals and Gems

21. Name specimens A and B.

22. A is 9 on the Mohs scale. Only one mineral is harder. Name it.

23. Name a gemstone that comes from A and a gemstone that comes from B.

24. B is often found as green grains inside which dark rock?

◯ A. granite ◯ B. basalt ◯ C. marble ◯ D. sandstone

Station 7: The Fizz Test

25. Put 1 drop of vinegar on A, B, and C and look with a hand lens. Which ones fizz?

26. Why do all three fizz?

27. Name specimens A, B, and C.

28. B and D are both limestone. How did each one form?

Station 8: Igneous Rocks

29. Name specimens A, B, C, and D.

30. Which specimen cooled the fastest? How can you tell?

31. Drop C into the cup of water. Does it float? Card D shows scoria: does it usually float or sink? What made the holes in both?

32. Where did all 4 of these rocks form?

◯ A. deep underground, cooling slowly ◯ B. at or near Earth's surface, from lava
◯ C. at the bottom of a lake ◯ D. under heat and pressure, without melting

Station 9: Rocks Made of Pieces

33. Name specimens A, B, and C.

34. Put A, B, and C in order from biggest grains to smallest grains.

35. Which specimen most likely formed from mud in very still water, like a lake bottom? Why?

36. Loose sediment turns into rock in 2 steps. Name them.

Station 10: Rock from Plants

37. Name specimen A. Is it igneous, sedimentary, or metamorphic?

38. What was A made from long ago?

39. A mineral must be formed in nature, not alive, solid, have its own chemical recipe, and have a crystal pattern. Is A a mineral? Explain.

40. Give one use of A and one problem it causes.

Station 11: Layered Metamorphic Rocks

41. Name specimens A, B, C, and D.

42. All 4 are foliated. What does foliated mean?

43. All 4 can form from the same parent rock as the heat and pressure keep rising. Name that parent rock.

44. Which specimen is used for roof tiles and old-time chalkboards?

Station 12: Sugary Metamorphic Rocks

45. Name specimens A and B.

46. Name the parent rock of A and of B.

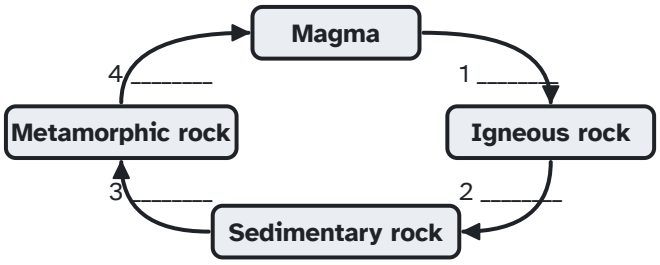
47. Do ONE test that proves which is which. What test did you do, and what happened?

48. Which word describes both A and B?

◯ A. foliated ◯ B. nonfoliated ◯ C. igneous ◯ D. sedimentary

Station 13: The Rock Cycle

Use the rock cycle diagram. Each number is an arrow.



49. A volcano erupts and the lava hardens into rock. Which numbered arrow is this, and what is the process called?

50. Rain and ice break a granite mountain into sand, and a river carries the sand away. Which arrow does this start? Name the 2 processes.

51. The diagram shows only 4 arrows. Can a metamorphic rock be broken into sediment? Explain.

52. What must happen to a metamorphic rock for it to become an igneous rock?

Station 14: Rocks and Minerals at Work

Answer from what you know or your binder. No specimens at this station.

53. Name the rock or mineral from our list used for each: (a) drywall, (b) kitchen countertops, (c) table salt, (d) glass and computer chips.

54. Which mineral on our list is crushed to make sandpaper because it is so hard?

55. Give 2 reasons minerals are important to a country's economy.

56. Feldspar, mica, and quartz are all mined in North Carolina. Which item is made using feldspar?

◯ A. pencils ◯ B. dishes, tiles, and other ceramics ◯ C. table salt ◯ D. electric wire

27	A calcite; B limestone (chemical); C limestone.
28	B: calcite came out of water and hardened (like in caves or springs). D: shells and skeletons of sea animals piled up and were pressed and cemented together.
Station 8: Igneous Rocks Coach: set out A basalt, B obsidian, C pumice, D the printed Scoria photo card, and a clear cup of water.	
29	A basalt; B obsidian; C pumice; D scoria.
30	B, obsidian. It is glass with no crystals at all.
31	C (pumice) floats; D (scoria) usually sinks. The holes were made by gas bubbles in the lava.
32	B. at or near Earth's surface, from lava
Station 9: Rocks Made of Pieces Coach: set out A conglomerate, B sandstone, C shale.	
33	A conglomerate; B sandstone; C shale.
34	A, B, C (pebbles, then sand, then mud).
35	C (shale). Tiny mud bits only settle out where the water is calm.
36	Compaction (pressing together) and cementation (gluing together).
Station 10: Rock from Plants Coach: set out A, the printed Bituminous coal photo card.	
37	Bituminous coal; sedimentary.
38	Swamp plants that were buried, pressed, and heated for millions of years.
39	No. It is a rock made from the remains of living things (plants) and has no crystal pattern.
40	Use: fuel to make electricity or steel. Problem: burning it makes air pollution and carbon dioxide (or mining can harm land and water).
Station 11: Layered Metamorphic Rocks Coach: set out A slate, B the printed Phyllite photo card, C schist, D gneiss.	
41	A slate; B phyllite; C schist; D gneiss.
42	The minerals are lined up in layers, sheets, or bands.
43	Shale. (Gneiss can also form from granite.)
44	A, slate.
Station 12: Sugary Metamorphic Rocks Coach: set out A marble, B quartzite, vinegar with a dropper, and a glass plate.	
45	A marble; B quartzite.
46	A marble: limestone. B quartzite: sandstone.
47	Vinegar: marble fizzes, quartzite does not. Or glass: quartzite scratches glass, marble does not.
48	B. nonfoliated
Station 13: The Rock Cycle	
49	Arrow 1; cooling (and hardening).
50	Arrow 2; weathering (breaking down) and erosion (carrying away).
51	Yes. Any rock at Earth's surface can be weathered and eroded into sediment; the rock cycle has many paths.
52	It must melt into magma, then cool and harden.
Station 14: Rocks and Minerals at Work	
53	(a) gypsum; (b) granite (marble or quartzite also OK); (c) halite; (d) quartz.
54	Corundum.
55	Mining and selling them gives people jobs and money; we need them to make metals, electronics, buildings, fuel, and many other products (any 2).
56	B. dishes, tiles, and other ceramics